ASSIGNMENT

TECHNOLOGY PARK MALAYSIA

CT124-3-2-MAE

MOBILE APP ENGINEERING

HAND OUT DATE: 09 DECEMBER 2025

HAND IN DATE: 08 FEBRUARY 2026

PROPOSAL

INSTRUCTIONS TO CANDIDATES:

1 Assignment is to be submitted through online submission (Moodle)

2 Students are advised to underpin their answers with the use of references (cited using the APA Referencing).

3 Late submission will be awarded zero (0) unless Extenuating Circumstances (EC) are uphelp.

4 Caes of plagiarism will be penalized.

5 Where the assignment should be submitted online (Moodle), the link to the source code and other relevant sources (where appropriate) should be attached together with the submission.

Table of Contents

Abstract 3
1.0 Introduction and Related Work 4
1.1 Introduction 4
1.2 Related Work 5
1.2.1 Feedme 5
1.2.2 TabSquare 6
2.0 Motivation 7
2.1 Problem Context 7
2.2 Limitations of Study 8
2.3 Novelty of Research Idea 9
3.0 Description of Process 10
3.1 Key Use Cases 10
3.2.1 UI & Wireframes 12
3.2.2 Architecture 20
3.3 Technical Requirements 21
3.3.1 Tools and Simulation Platforms Used 21
3.3.3 Third-Party API Integration 24
4.0 Gantt Chart 26
5.0 References 27
6.0 Appendix 28
6.1 Work Breakdown Structure 38

Abstract

Recently, the Food and Beverage (F&B) industry in Malaysia has been under a lot of pressure to optimize operations and improve customer satisfaction during the busiest hours. However, most traditional outlets still have to overcome major problems when it comes to their manual ordering and billing systems. On top of that, employees suffer from ordering communication problems because they don't have a single digital platform to work with. When restaurants lack of tools to keep track of their staffs' orders and inventory in real-time, it naturally demands a lot of effort from them to deliver great services.

This paper first highlights the introduction and related work of the Smart F&B QR Ordering System. After that, the motivation part talks about the problem context, study limitations, and the novelty of the proposed research ideas.
Next, it lays down the primary scenarios of use for each user segment: Customers, Service and Kitchen Staff, and Restaurant Managers, also the planned system's design methods and technical requirements. At last, the Gantt chart summarizing the schedule for the application development project will be displayed.

1.0 Introduction and Related Work

1.1 Introduction

Food and Beverage (F&B) industry has undergone a pronounced digital shift over the past years, in accordance to the demand to increase the speed, convenience, and contactless service. The conventional restaurant business methods that depend on orders handwritten, manual billing, and verbal communication are fast becoming inefficient in a busy restaurant setting. The current customers are demanding smooth digital experiences just like those experienced in online e-commerce sites. The generative research carried out in this project demonstrated that the diners are more concerned with time efficiency, the visual representation of the menu, and painless payment systems. Lots of participants complained about time wasting when waiting to get bills, the inability to divide payment and the inability to know the status of orders. Such problems have adverse effects on customer satisfaction, and could affect the choice of the restaurant. In terms of operation, manual workflows cause communication barriers between front of house employees and kitchen employees which results in inaccuracies in orders and prevents food wastage. The restaurant managers have also been constrained in having real-time access to sales, inventory and operational performance, which inhibits them in coming up with timely decisions regarding their businesses. In order to overcome these issues, the current project suggests the creation of a Smart F&B QR Ordering System, which is a cross-platform mobile application that would connect diners, service workers, kitchen employees, as well as managers into a single digital environment. The system will be implemented to facilitate the ease of ordering, promote immediate communication, automate the billing, and offer a centralized managerial control. The proposed solution is aimed at enhancing the customer experience, increasing operational effectiveness, and data-driven decision-making by using mobile technology in the modern restaurant setting.

1.2 Related Work

1.2.1 Feedme

FeedMe is an online ordering system that allows clients to order food and pay with QR codes after reading through menus. The system makes dining more efficient by reducing the number of staff that a customer needs to interact with and the waiting time is also minimized. FeedMe has a big strength in the form of its user-friendly interface where customers can control their dining experience on their own. Also, the digital payments facilitate the ease in the financial transactions and minimize human error. Nonetheless, FeedMe is more of a customer-side ordering platform, and does not cover more thoroughly the operational issues like real-time kitchen coordination, automatic updates on inventory, or managerial analytics centrally. Even in the absence of powerful backend integration, restaurants can still have communication breakdowns and operational blind spots.

1.2.2 TabSquare

TabSquare is a smart restaurant application that integrates digital menu, self-ordering system and data analytics. It favors interactive menu displays including pictures and suggestions, which assist the customer in choosing wisely as well as boosting restaurant sales. One of the strongest features of TabSquare is that it is analytics-based and helps restaurants to learn their customers and manage the menu to better suit customer preferences. The digital transmission of orders is another way the system enhances accuracy of order. Although they have their strengths, TabSquare in many cases demands specialized hardware including tablets, thus implementation is very expensive to small and medium size restaurants. In addition, it lacks flexibility to mobile-based solutions because it relies on fixed devices.

Research Gap

Even though the current systems can enhance digital ordering, most of them do not offer an entirely integrated ecosystem to link customers, kitchen employees, and managers in real time using a single mobile platform.
To fill this gap, the proposed Smart F&B QR Ordering System will provide:

- Cross-role synchronization
- Live inventory visibility
- Real-time order tracking
- Automated billing
- Focal point management dashboard.

This integrated solution makes the system a scalable and low cost solution to the contemporary restaurant.

2.0 Motivation

2.1 Problem Context

The generative study has found out a number of inefficiencies in the conventional restaurant processes that had a detrimental impact on customer experience and performance levels. On the side of the customer, it was observed that manual billing was a time-consuming activity especially whereby the diners had to wait to be given receipts or dividing bills between groups. It was also reported that participants had problems in understanding menus that were presented in text only format and that they preferred text representations that assisted them in making decisions more quickly. As well, there are no real-time updates on orders, which cause uncertainty about waiting time, which also leads to dining anxiety. One of the other operational problems could also be seen in the service and kitchen staff as the use of hand written orders and verbal communication often caused wrong orders, missed special orders and wastage of food. Employees complained that they were physically tired due to the constant movement between the tables and kitchen to pass the information. At the management level, the inability to have a centralized view of operations means that the restaurant owners are unable to easily track the sales, manage the inventory and identify financial variances. The manual reconciliation processes are time consuming as well as subject to human error. These results demonstrate the necessity of a single digital solution that can increase transparency, boost the communication process, and automate the most important processes in the restaurant.

2.2 Limitations of Study

1. Insecure Payment Gateway Integration:

Even though third-party APIs such as Stripe and local E-wallet providers were technically available for real-world transaction integration, the student project fundamentally lacks the capability to securely process real-time refunds and handle sensitive banking data. As a result, this constitutes more than just a simple obstacle - it is a critical one. This limitation entails the risk of loss of money and the erosion of trust from the users. In the absence of a fully secure and certified payment gateway, the app is not in a position to conduct actual monetary transactions on its own in this particular stage of development.

2. Focus on Counter-Based Payments:

The system has been designed to mainly facilitate counter-based payments via QR verification at this stage to ensure both the security and the simplicity of the present prototype. Considerable changes were made in the system to accommodate the traditional operation mode of the F&B industry as revealed by the interviews, which, however, encumber the "Pay & Go" experience to a physical interaction at the counter thus negating an entirely remote checkout option.

3. Sample Size and Testing Constraints:

Only eleven participants from three principal user categories: Customers, Service/Kitchen Staff, and Restaurant Managers have been involved in the study phase. Therefore, the results of the study cannot be generalized to the overall Malaysian F&B industry population due to the small experimental sample. Although the discovery of qualitative manual bottleneck insight proved to be immensely helpful, the limited pieces of information obtained may fail to strongly represent the reality of various regional operational practices or different restaurant types in Malaysia.

4. Device-Specific Feature Limitations:

The app development has been done with the use of Flutter to remain compatible with multiple platforms; however, features such as live Kitchen Display System (KDS) sync and order follow-up notifications are not yet fully optimized for a large variety of devices. The experiments have been mainly performed on Android-based devices , which means that there might be potential issues with the user interface and/or user experience or performance delays if the app is run on older models or in changing network conditions typical of a crowded restaurant space.

2.3 Novelty of Research Idea

The innovation of the Smart F & B QR Ordering System comes from its capability to be a complete, role-specific, operational ecosystem that merges all major facets of the dining experience into one mobile app. Most modern mobile apps at present only provide single features such as a digital menu, delivery service, or point-of-sale systems, separately. However, this system stands out as it integrates these features into a harmonious digital lifecycle that is specifically tailored for the Malaysian F & B industry.

Apart from the multiple user role inclusions, the system separates each role—Customer, Staff, and Manager—in terms of dedicated functional modules within a single integrated system. Without having to switch between different apps or fragmented communication platforms, the system offers a one-stop environment where all ordering, kitchen coordination, and business management tasks can be done collaboratively and in real time.

- Customers are not just given a way to browse through food items via the platform. Personalized time and social convenience management is further supported by the real-time transparency features of the platform, such as live order tracking (e.g., "Cooking," "Serving") and automated "Go-Dutch" bill splitting. Such features encourage diner time management and social comfort, which are usually neglected in general delivery or reservation apps.

- If you are a member of the Service and Kitchen Staff, your role will be completely changed by the system as you will no longer be a mere recipient of orders, but a manager of workflows. Verbal instructions that are not reliable and orders written on paper that can be hardly read are now replaced by a coordinated Kitchen Display System (KDS) that can be viewed from the Kitchen area. Besides, you can check your live inventory, send out instant "Sold Out" messages, and handle each stage of the production process through the platform, all of which demonstrate the type of tasks involved in a busy kitchen operation.

- Restaurant Managers can enjoy features that are not commonly available in the usual apps of their kind. For instance, they can keep an eye on their restaurant from a distance, control the pricing of dishes centrally, and finalize the financial records of the day automatically. These kinds of logistics features are just not more efficient but also quite tailored to Malaysian restaurants' working settings to the point that they not only guarantee financial integrity but also help to curtail the incidence of internal theft.

In short, the Smart F & B QR Ordering System is far more than a digital menu, it is a digitally integrated operation platform that covers all phases. Functionality that has lately been scattered over lottery chits, whiteboards for inventories, and manual accounting books is thus centralized by it. Mainstream applications rarely exhibit this level of industry targeting and full-cycle process management which is what characterizes this system as an original and handy answer to the disjointed state of restaurant operations in Malaysia.

3.0 Description of Process

3.1 Key Use Cases

According to the pain points that were defined at the generative research stage, including the ineffectiveness of the handwritten orders and the inability to track the inventory in real-time, the suggested Smart F&B QR Ordering System targets three key user groups, including Customers (Diners), Service and Kitchen Staff, and Restaurant Managers.

3.1.1 Customers (Diners)

The main ambition of the customers is to have a smooth Pay and Go dining experience that removes manual friction. (Citing Research Insight: "Diners expect a stress free experience and visual menus to make it happen.)

- **Use Case 1: Scan & Visual Menu Browsing**
  - A table-dependent QR code can be scanned by users to get a digital menu where images are featured. This is a solution to the text-only menu pain point which offers high-quality photos and bilingual descriptions to assist the decision-making process.
- **Use Case 2: Digital Order & Customization**
  - The items can be selected, preferences can be customized (e.g. No Spicy, Less Sugar), and orders can be placed through the device itself, waste time of the staff members on taking orders manually is minimized.
- **Use Case 3: Split Bill Payment**
  - This feature can be used to resolve the frustration associated with the manual calculation of the bill, which has been determined in the research, since it may enable splitting the bill per item or equally among the group and make direct payments using an unique order number and QR Code at the counter

3.1.2 Service and Kitchen Staff

The main aim of the personnel is to minimize physical exhaustion and communication mistakes between the front of the house and back of the house. (Citing Research Insight: "Staff use synchronized digital display to substitute sudden manual communication).

- **Use Case 1: Kitchen Display System (KDS) View**
  - Kitchen employees have a landscape KDS function showing incoming orders with large fonts and color tags (e.g., Allergies), so any human error of the illegibility of handwritten remarks (on paper chits) is removed.
- **Use Case 2: One-Tap "Ready" Notification**
  - Chefs do not need to shout in a crowded kitchen, as a tap on the KDS Ready button will immediately send a silent signal to the device of the waiter, letting him or her collect and serve the food.
- **Use Case 3: Real-Time Status Updates**
  - Waiters will be able to see the current status of an order of any table (e.g., Cooking, Ready) on their mobile device, and will be able to respond to the questions of the customers without running to the kitchen.

3.1.3 Restaurant Manager

The major objective of managers is to achieve remote operation visibility and centralized control of the business. (Citing Research Insight: "Managers require centralized, digital control... to stop false orders...).

- **Use Case 1: Remote Sales Dashboard**
  - Managers will have access to real-time sales data, table turnover rates, and revenue reports regardless of location using a mobile or web dashboard, overcoming the problem of having a blind spot since they are off-premises.
- **Use Case 2: Instant Inventory Toggle (Out-of-Stock)**
  - Managers can put an item as being unavailable immediately in all customer menus with a single tap. This allows customers to not have to order sold-out items, and service recovery friction is avoided.
- **Use Case 3: Centralized Menu Management**
  - Enables managers to recalculate prices or make amendments to menu descriptions over or multiple outlets in bulk eliminating the need to physically make changes (e.g. stickers on menus).

3.2 Design Strategies

3.2.1 UI & Wireframes

The User Interface (UI) design is concerned with accessibility, cultural relevance, and user friendly. The design system is designed in such a way that it reduces the learning curve of a wide range of users including tech savvy students and the older restaurant employees.

1. App Identity & Logo

Figure 3.2.1.1: *App Logo*

App Name: Makan Je

It is named after the Malaysian colloquialism Makan Sahaja (Just Eat). The name Je is the symbol of the philosophy of the system, Simplicity. To Customers, it pledges to offer them an easy-going dining experience where they can simply eat without any concerns of waiting in lines or making payments. To Staff and Managers**,** It is an indication of an efficient workflow with sophisticated manual processes automated. The plate icon is the most important feature of the logo and immediately identifies the application as the F&B sector. The design is simple and easygoing, which can be associated with no-stress promises of the brand.

2. Color Palette

The color strategy is selected considering the color psychology in the food service industry.

- **Main Color: Orange**
  - Orange is a well-established brand in the F&B sector since it has been known to stimulate appetite and stimulate impulse orders. Moreover, it also exudes a feeling of energy, swiftness and rush-paced service, and this fits the objective of the system which is to shorten wait times.
- **Secondary/Neutral Colors: White & Dark Grey**
  - These two colors are used in the backgrounds and text to have a clean interface. The Kitchen Display System (KDS) should be provided with high-contrast text (Dark Grey on White) to be readable.

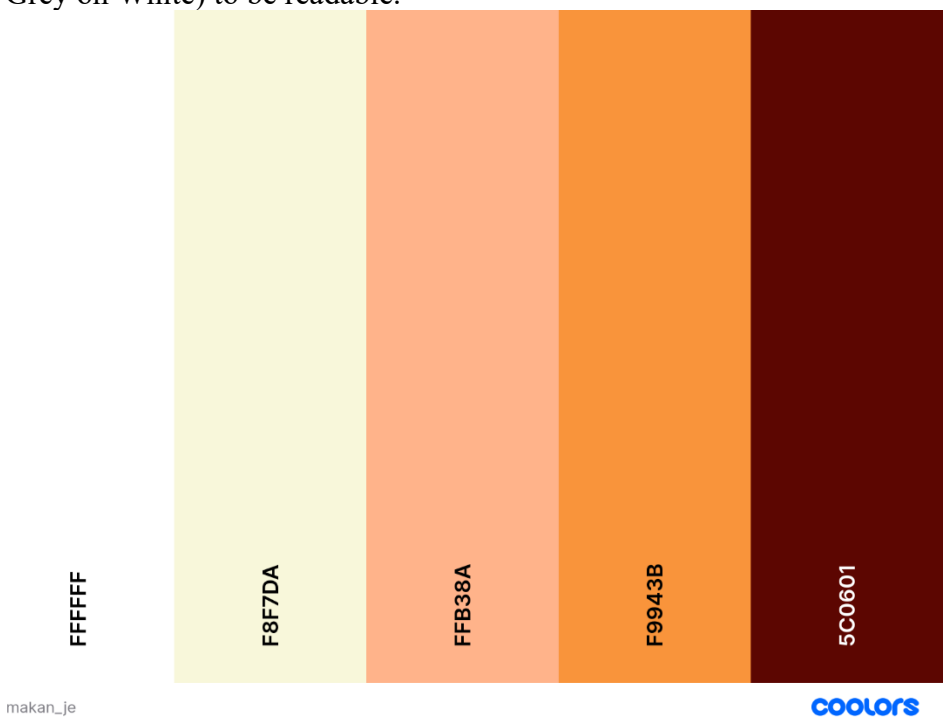

Figure 3.2.1.2: *Color Palette*

3. Typography

The system has a classic Sans-Serif typeface (e.g., Roboto or Open Sans) to make sure that it is as legible as possible on various devices (including smartphones of customers and tablets in the kitchen).

We did not use decorative or too stylized fonts. Functional clarity is the priority, so that the chefs are able to read the orders even when they are a long distance away, and the customers do not experience visual exhaustion when navigating the menu.

3. Wireframe

A. Customer Interface (The "Makan Je" Experience)

Digital Visual Menu

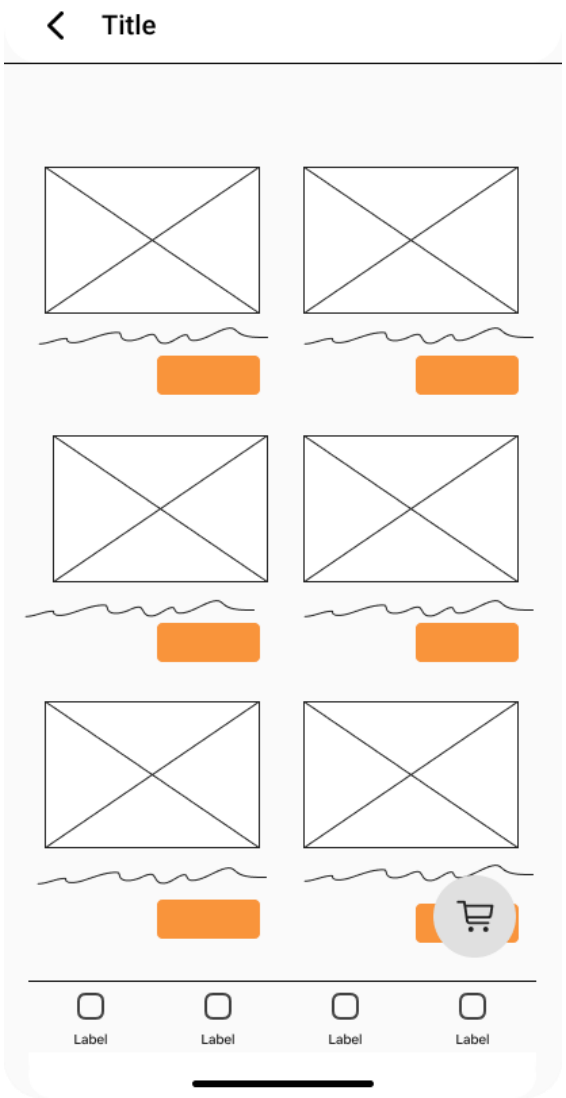

Figure 3.2.1.3: Customer Visual Menu

In order to eliminate the issue of language barrier and absence of visual data that was found in the study, the customer interface focuses on visual-grid layout. The food imagery is rich in content, and it occupies 60 percent of the screen space to help in the decision-making process, but the Add to Order buttons will be represented in the main color, Orange, to induce impulse buying.

Split Bill Payment

Figure 3.2.1.4: Split Bill Feature

This interface addresses social friction of manual calculation when a customer arrives with a large number of people. Before the user can move on to the payment gateway, they can easily select to split equally or pay per item, and this ensures a frictionless Pay and Go experience.

B. Service & Kitchen Staff Interface (The Workflow)

Kitchen Display System (KDS)

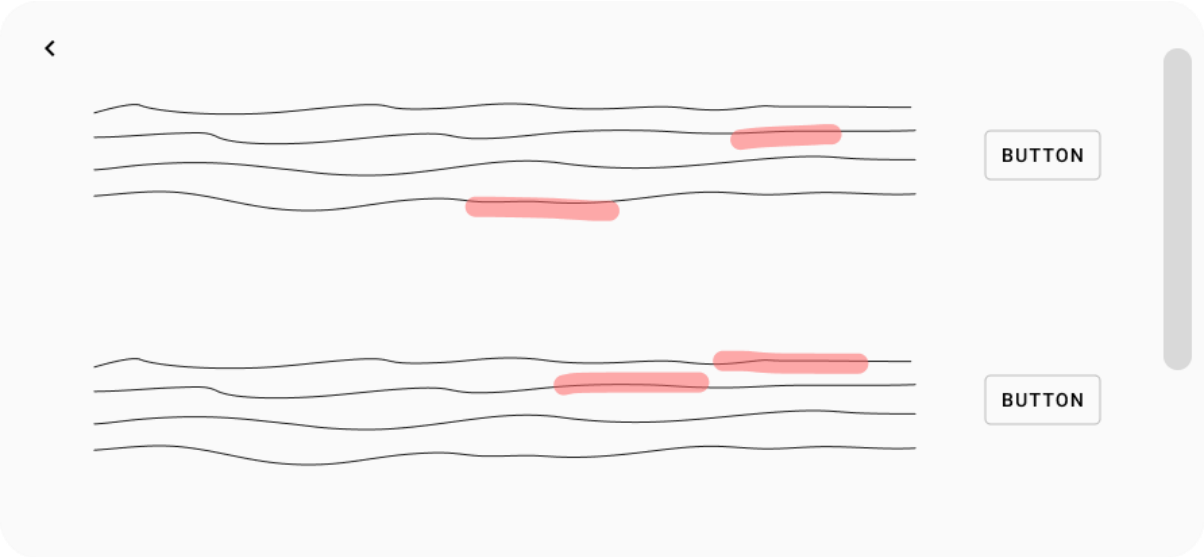

Figure 3.2.1.5: Kitchen Display System (KDS)

The KDS is designed to be used in high-stress and noisy environments. Typography is greatly exaggerated to enable one to read it even at a great distance. Important custom comments (e.g. allergies) are emphasized in Red to avoid cooking mistakes, which is better than the unreliable handwritten chits.

Waiter Notification

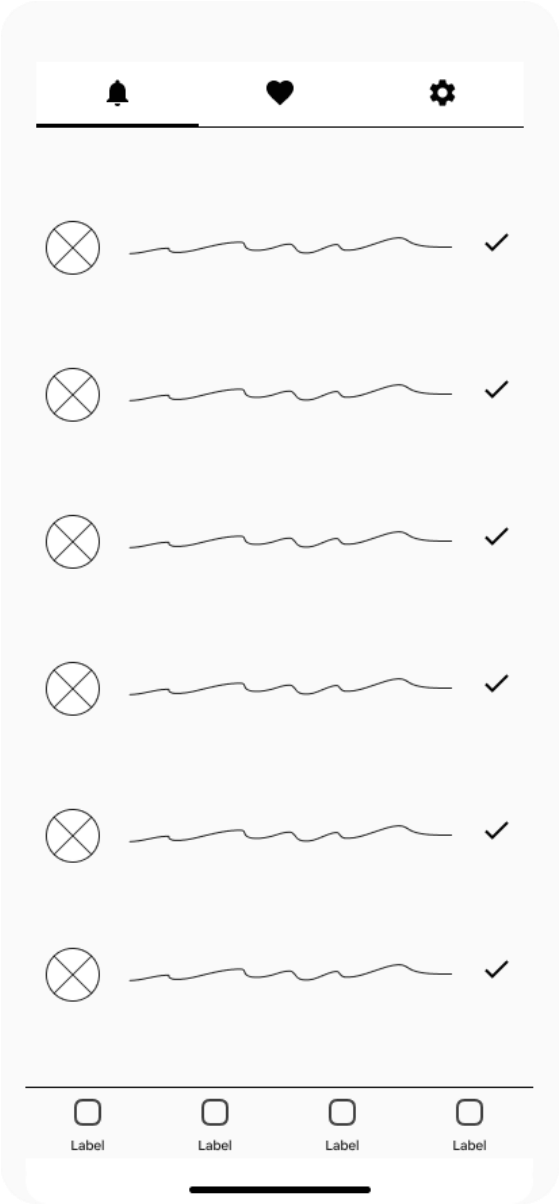

Figure 3.2.1.6: Real-time "Ready" Notification

Rather than using shout, this interface establishes a non-verbal online connection between kitchen and floor employees. The KDS sends instant push notifications to waiters when food has been marked as ready on the KDS, so food is served hot.

C. Restaurant Manager Interface (The Control Center)

Manager Dashboard

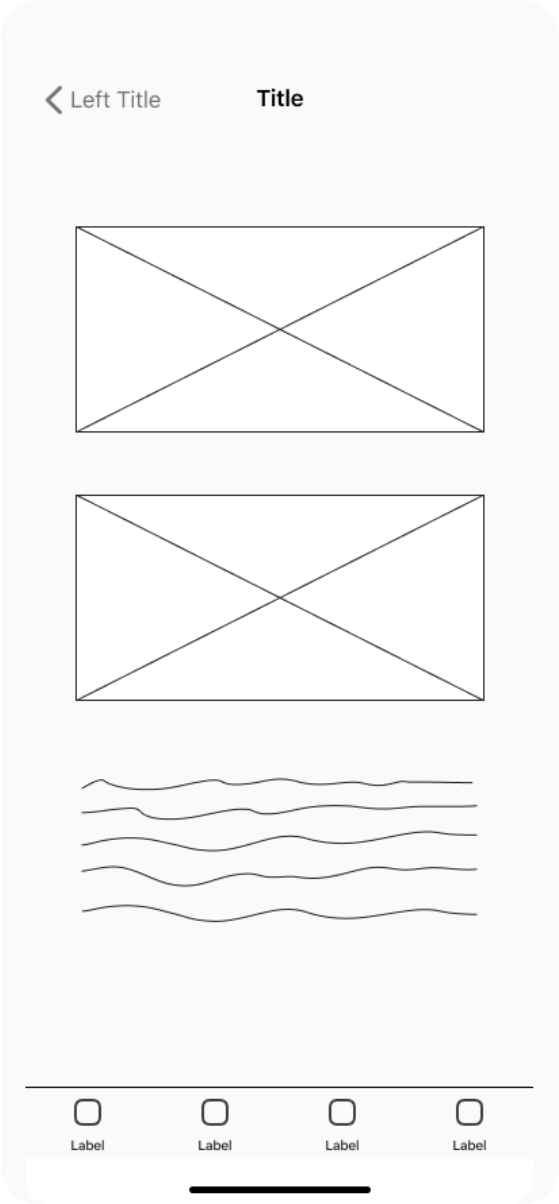

Figure 3.2.1.7: Remote Manager Dashboard

This dashboard will resolve the problem of the blind spot among owners. It will also visualize live sales and best selling items so that managers can monitor the business performance even when they are not present in the outlet.

Menu Availability Toggle

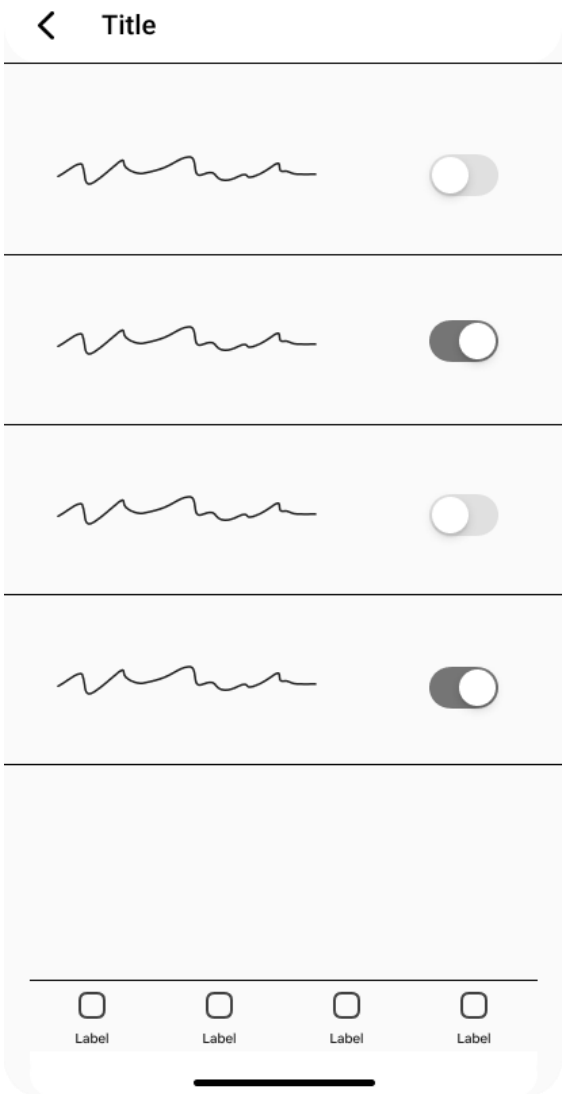

Figure 3.2.1.8: Instant Inventory Toggle

This gives the manager the ability to change the out of stock status of all the customer devices at once by a single toggle. This eliminates the scenarios of false order that were found in the study where customers place orders on items that are out of stock.

3.2.2 Architecture

1. MVVM Architecture

To achieve separation of concerns between the UI, business logic, and data management layers, the application development leveraged the Model-View-ViewModel (MVVM) pattern. This architectural style encourages decoupling of different parts/components so that one doesn't have to worry about another. Hence, even if one module (say, Kitchen Display module) is broken, it can be fixed (debugged) without interfering with the functioning of other parts, thus, increasing maintainability, scalability, and development efficiency.

Below is the file structure of the Flutter source code:

```
lib/                          // Root directory of the Flutter application code.
├─ core/                      // Contains shared utilities and app-wide settings.
│  ├─ constants/              // Stores global keys, URLs, and asset paths.
│  ├─ locale/                 // Manages localization for English, Malay, and Chinese.
│  ├─ theme/                  // Defines app colors, fonts, and UI styles.
│  └─ utils/                  // Helper functions for dates, currency, etc.
├─ data/                      // Handles all data operations and backend communication.
│  ├─ providers/              // Manages real-time data streams (e.g., from Firebase).
│  ├─ repositories/           // Abstraction layer for fetching and caching data.
│  └─ services/               // Direct interactions with external APIs (e.g., Payment).
├─ models/                    // Defines the data structures (Plain Old Dart Objects).
│     order_model.dart        // Blueprint for order details and status.
│     menu_item_model.dart    // Blueprint for food items, prices, and image URLs.
│     user_model.dart         // Blueprint for different user roles (Customer, Staff, Manager).
├─ view_models/               // Contains the business logic for the UI (MVVM).
│     customer_view_model.dart // Manages cart, ordering, and bill-splitting logic.
│     staff_view_model.dart   // Handles KDS logic and inventory updates.
│     manager_view_model.dart // Processes sales data for the dashboard.
├─ views/                     // Contains the UI screens and widgets.
│  ├─ common/                 // Shared screens like Login and Splash.
│  ├─ customer/               // Screens specifically for diners.
│  │     menu_view.dart       // The main visual QR menu screen.
│  │     tracking_view.dart   // Screen showing live order progress.
│  │     payment_view.dart    // Screen for payment and bill splitting.
│  ├─ staff/                  // Screens for kitchen and service staff.
│  │     kds_view.dart        // The Kitchen Display System screen.
│  │     inventory_view.dart  // Screen to manage "Sold Out" items.
│  └─ manager/                // Screens for restaurant managers.
│        dashboard_view.dart  // The main sales monitoring dashboard.
│        menu_mgmt_view.dart  // Screen for updating menus and prices.
└─ widgets/                   // Reusable UI components used across different screens.
      custom_buttons.dart     // Standardized button styles.
      order_card.dart         // A common card widget to display order details.
      loading_indicator.dart  // A shared loading spinner widget.
```

Figure 3.2.2.1: Final Proposed File Structure of Flutter Source Code

3.3 Technical Requirements

3.3.1 Tools and Simulation Platforms Used

1. Visual Studio Code

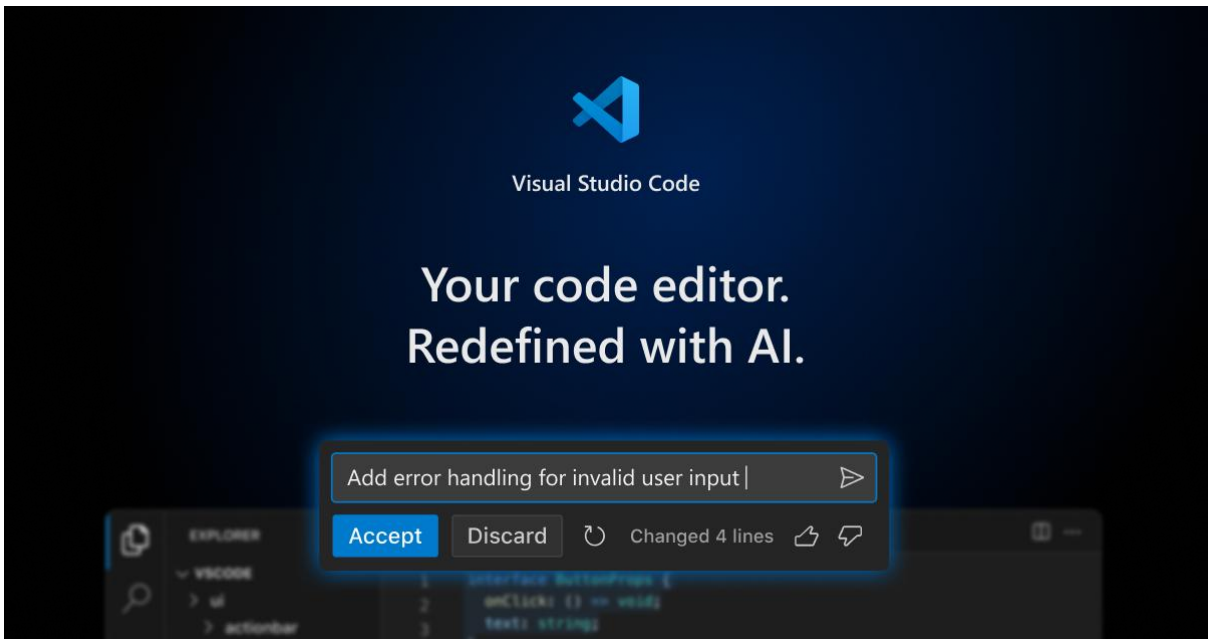

Figure 3.3.1.1: *Visual Studio Code Logo*

As the main development tool for the Smart F&B QR Ordering System, Visual Studio Code, a lightweight and open-source IDE, is used. It is chosen mainly because the tool has several useful extensions like Flutter and Dart support, Git integration, and intelligent code completion as well as error checking features. Hence, the development team is enabled to work productively and the system development and maintenance are smooth.

2. Android Studio

Figure 3.3.1.2: *Android Studio Logo*

Android Studio can simulate Android devices right inside a desktop environment. It comes with the Android Software Development Kit (SDK) and other tools, which allow the development team to establish a connection with both virtual and real Android devices, test and debug applications on them. This is very important, especially when testing the live Kitchen Display System (KDS) and customer tracking interfaces in a situation as close as possible to real operational conditions.

3.3.2 Backend and Storage

1. Firebase

Figure 3.3.2.4: *Firebase Logo*

Firebase is a cloud-based platform from Google that offers a set of backend services for application development. The main reason to use Firebase as the backend and storage for the Smart F&B QR Ordering System was its excellent compatibility with Flutter, thus making the development process easier and faster. In addition to that, it is a low-cost solution that also offers many great features including real-time database, user authentication, and cloud storage.

It is a pivotal move because the developers can leverage the platform to secure user records and at the same time, backend services can be kept scalable with barely any configuration and upkeep. To be exact, the system extensively relies on Cloud Firestore to accomplish all the tasks involving the flow of live data between devices like updating and reflecting the current status of orders ("Cooking," "Served") or the availability of items, instantly, everywhere. This solution allows, for example, when the manager or the staff mark a particular dish as "Sold Out," all customers will immediately see the change on their menus, and thanks to this, there will be no invalid orders and the staff will not have to apologize for that.

3.3.3 Third-Party API Integration

1. Google Maps API

Figure 3.3.3.1: *Google Maps API Logo*

The integration of Google Maps API serves the purpose of giving precise location visualization and routing functionalities in the Smart F&B QR Ordering System.

This kind of integration becomes highly significant when it is through it that restaurant managers and owners of chains can pinpoint their branches' locations and, at the same time, determine the area spread of their customers. Google Maps API has been chosen mainly because of its up-to-date data and very high reliability, thus guaranteeing users accurate and up-to-date geographical information whenever they plan their operations or coordinate their deliveries.

2. QR Flutter Library (Internal Logic)

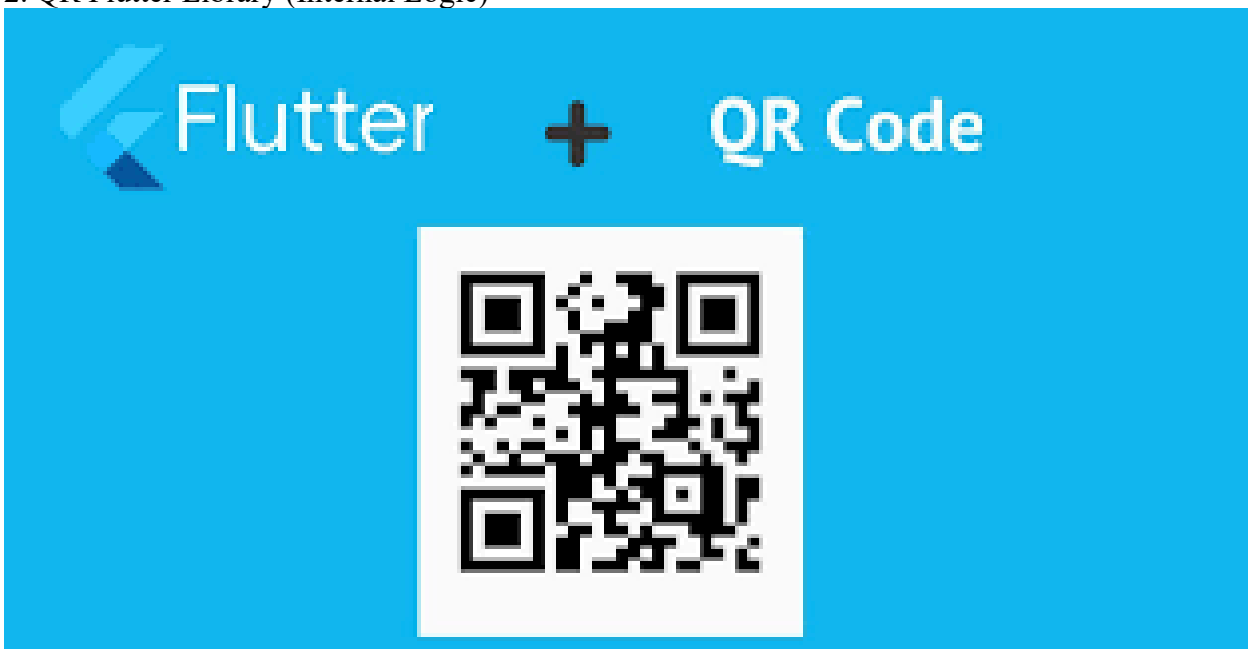

Figure 3.3.3.2: *QR Code Generation Logic*

Instead of complicated online payment gateways, the system makes use of the qr_flutter library to create a unique Digital Receipt QR for each order. This method is critical for supporting the "Pay at Counter" workflow, which is a good fit for traditional restaurant owners who want to keep the physical cash register under their control.

When a customer has finished their meal, the app creates a QR code with their unique Order ID and a "Split Bill" total that has been calculated. The customer only needs to show this code at the counter for the cashier to scan. This solution greatly lowers the "social friction" of manual math at the register and eliminates mistakes in bill-splitting, which were major issues according to the generative research. This way, the end-of-day account balance will coincide with the digital records without the need for sensitive banking integrations.

4.0 Gantt Chart

Figure 4.0.1: *Project Gantt Chart*

5.0 References

Bakar, A. S., & Phebe, L. (2025). Consumer perception and satisfaction of Quick Response (QR) code in casual dining restaurants in Malaysia. *Quantum Journal of Social Sciences and Humanities*, *6*(5), 45-56.

Ma, X., Li, Y., & Chen, H. (2025). As tasty as it looks: Exploring the impact of image aesthetics on food delivery sales. *Journal of Electronic Commerce Research*, *26*(2), 119-138.

Çiçek, B., & Doğan, M. (2025). The impact of sustainable QR menus on service quality and customer satisfaction: The moderating role of perceived risk. *Sustainability*, *17*(5), 2323. https://doi.org/10.3390/su17052323

Miswan, A. F., Ismail, M. N. I., & Zulkifly, M. I. (2025). Enhancing customer loyalty through QR code menu system: The satisfaction in urban Malaysian restaurants. *Quantum Journal of Social Sciences and Humanities*, *6*(S13), 32-46.

Kumar, P., & Singh, R. (2025). Cloud-based smart food service inventory management systems: Business applications and innovations. In *Digital Transformation in the Food Service Industry* (pp. 88-102). IGI Global. https://doi.org/10.4018/978-1-6684-1234-5

6.0 Appendix

Figure 3.2.1.1: *App Logo*

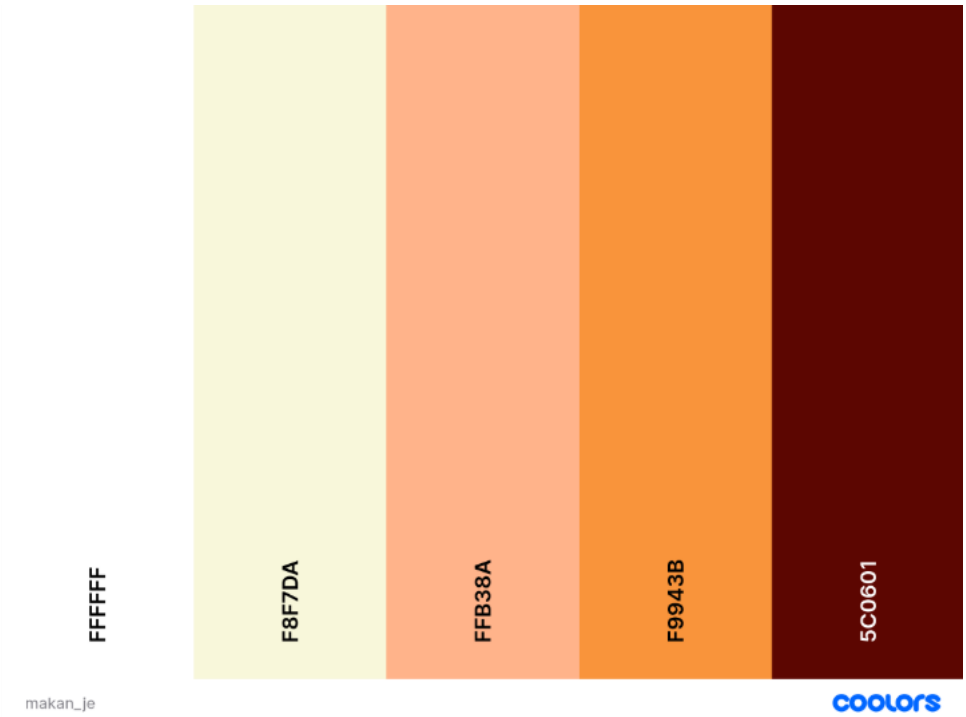

Figure 3.2.1.2: *Color Palette*

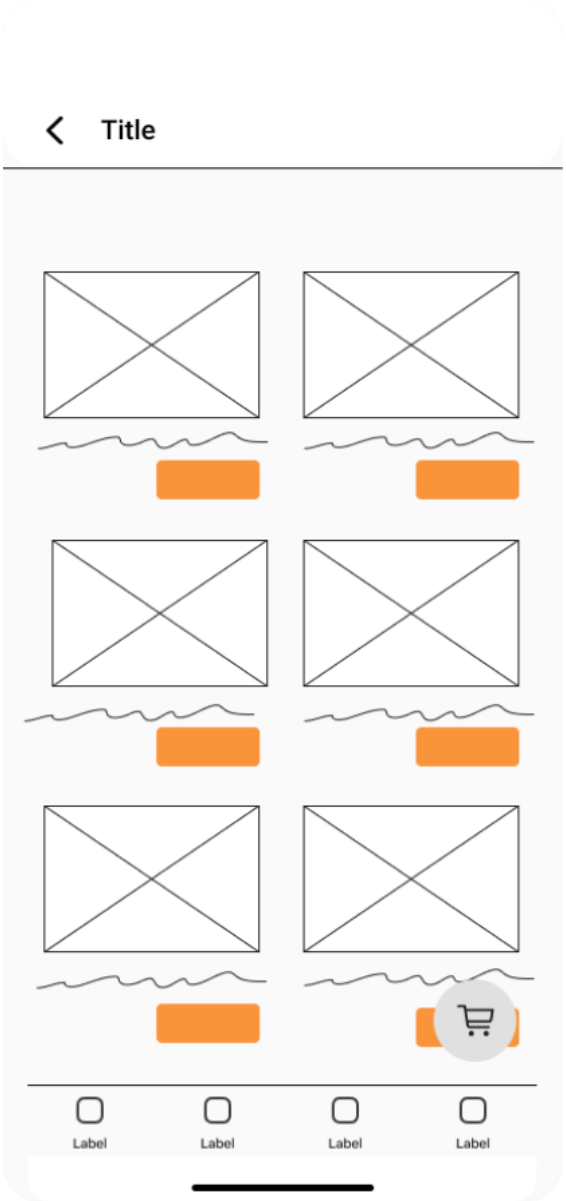

Figure 3.2.1.3: Customer Visual Menu

Figure 3.2.1.4: Split Bill Feature

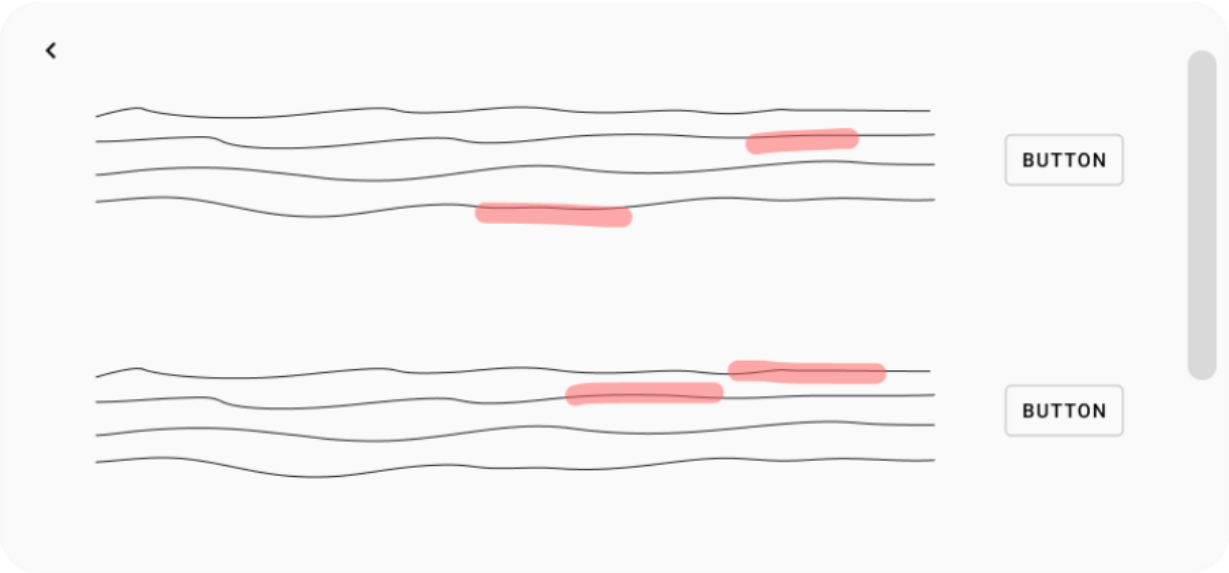

Figure 3.2.1.5: Kitchen Display System (KDS)

Figure 3.2.1.6: Real-time "Ready" Notification

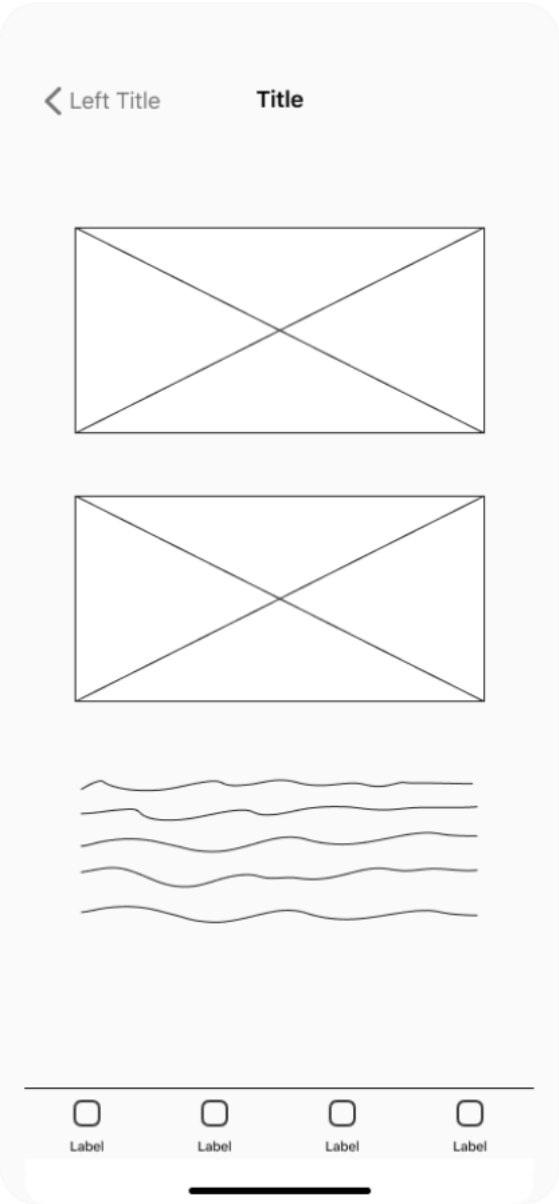

Figure 3.2.1.7: Remote Manager Dashboard

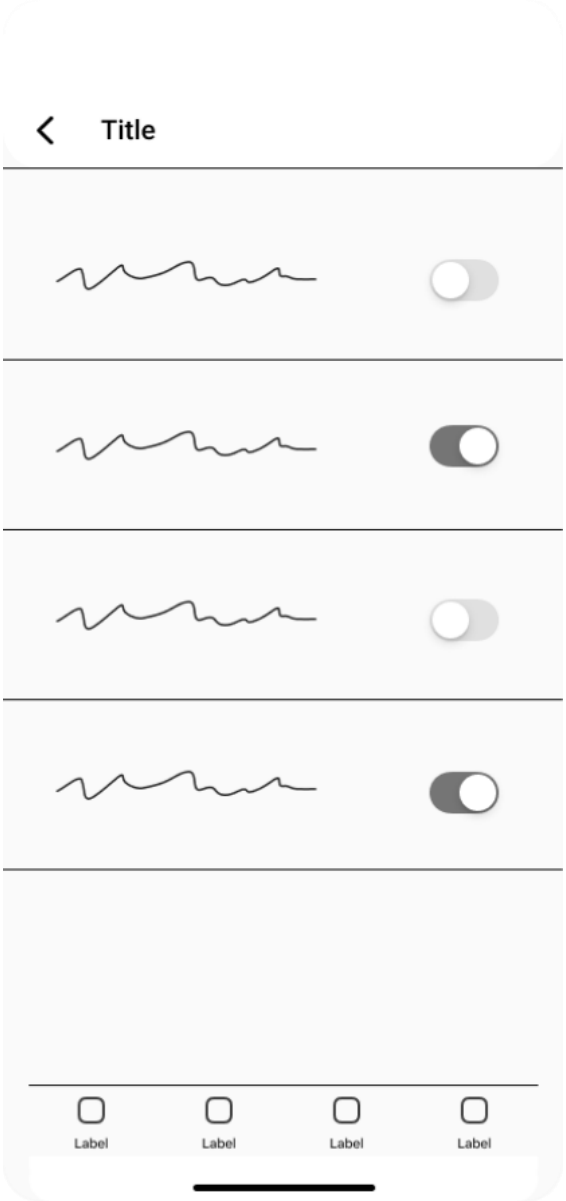

Figure 3.2.1.8: Instant Inventory Toggle

```
lib/                          // Root directory of the Flutter application code.
├─ core/                      // Contains shared utilities and app-wide settings.
│  ├─ constants/              // Stores global keys, URLs, and asset paths.
│  ├─ locale/                 // Manages localization for English, Malay, and Chinese.
│  ├─ theme/                  // Defines app colors, fonts, and UI styles.
│  └─ utils/                  // Helper functions for dates, currency, etc.
├─ data/                      // Handles all data operations and backend communication.
│  ├─ providers/              // Manages real-time data streams (e.g., from Firebase).
│  ├─ repositories/           // Abstraction layer for fetching and caching data.
│  └─ services/               // Direct interactions with external APIs (e.g., Payment).
├─ models/                    // Defines the data structures (Plain Old Dart Objects).
│  ├ order_model.dart         // Blueprint for order details and status.
│  ├ menu_item_model.dart     // Blueprint for food items, prices, and image URLs.
│  ├ user_model.dart          // Blueprint for different user roles (Customer, Staff, Manager).
├─ view_models/               // Contains the business logic for the UI (MVVM).
│  ├ customer_view_model.dart // Manages cart, ordering, and bill-splitting logic.
│  ├ staff_view_model.dart    // Handles KDS logic and inventory updates.
│  ├ manager_view_model.dart  // Processes sales data for the dashboard.
├─ views/                     // Contains the UI screens and widgets.
│  ├─ common/                 // Shared screens like Login and Splash.
│  ├─ customer/               // Screens specifically for diners.
│  │  ├ menu_view.dart        // The main visual QR menu screen.
│  │  ├ tracking_view.dart    // Screen showing live order progress.
│  │  ├ payment_view.dart     // Screen for payment and bill splitting.
│  ├─ staff/                  // Screens for kitchen and service staff.
│  │  ├ kds_view.dart         // The Kitchen Display System screen.
│  │  ├ inventory_view.dart   // Screen to manage "Sold Out" items.
│  └─ manager/                // Screens for restaurant managers.
│     ├ dashboard_view.dart   // The main sales monitoring dashboard.
│     ├ menu_mgmt_view.dart   // Screen for updating menus and prices.
└─ widgets/                   // Reusable UI components used across different screens.
   ├ custom_buttons.dart      // Standardized button styles.
   ├ order_card.dart          // A common card widget to display order details.
   ├ loading_indicator.dart   // A shared loading spinner widget.
```

Figure 3.2.2.1: Final Proposed File Structure of Flutter Source Code

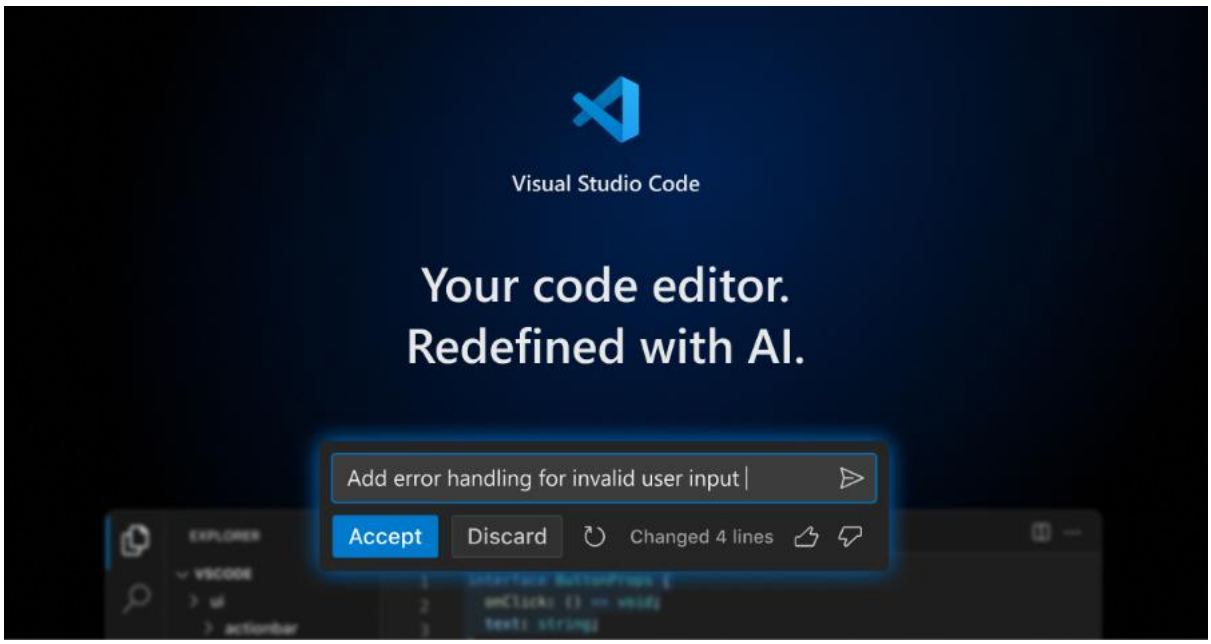

Figure 3.3.1.1: *Visual Studio Code Logo*

Figure 3.3.1.2: *Android Studio Logo*

Figure 3.3.2.4: *Firebase Logo*

Figure 3.3.3.1: *Google Maps API Logo*

Figure 3.3.3.2: *QR Code Generation Logic*

6.1 Work Breakdown Structure

WBS Code	Task Name / Description	Assigned To	Task %	Parent %
1.0	**Abstract & Introduction Phase**			**15%**
1.1	**Abstract:** Write a 150-word summary of the entire project (Problem + Solution).	Zi Yee	4%	
1.2	**1.1 Introduction:** Introduction to F&B digital transformation trends.	Ayaan	3%	
1.3	**1.2 Related Work:** Compare 2 competitors (e.g., Feedme, TabSquare) and highlight what they lack.	Ayaan	8%	
2.0	**Project Motivation & Context**			**13%**
2.1	**2.1 Problem Context:** Summarize pain points found in Part 1 to justify the need for this system.	Ayaan	5%	
2.2	**2.2 Limitations:** List constraints (e.g., Simulated Payment, Android-only testing).	Zi Yee	3%	
2.3	**2.3 Novelty of Research Idea:** Explain why the "All-in-One Role-Based Ecosystem" is unique.	Zi Yee	5%	
3.0	**System Description & Design (Core Technical Work)**			**55%**
3.1	**Functional Requirements**			**13%**
3.1.1	**3.1 Key Use Cases:** List detailed functions for Customer, Staff, and	Jin Jie	13%	

	Manager roles (e.g., "Place Order", "Update Stock").			
3.2	**System Design**			**37%**
3.2.1	**3.2.1 UI & Wireframes:** Create Logo, Color Palette, and 9+ Wireframes (Low-fidelity sketches for all 3 roles).	Jin Jie	25%	
3.2.2	**3.2.2 Architecture:** Create the MVVM Architecture Diagram and explain the Folder Structure.	Zi Yee	12%	
3.3	**Technical Implementation Details**			**5%**
3.3.1	**3.3 Technical Requirements:** Detail the tech stack: Flutter, Firebase (Auth/Firestore), and Google Maps API.	Zi Yee	5%	
4.0	**Project Management & Planning**			**8%**
4.1	**4.0 Gantt Chart:** Create the project timeline spanning Week 1 - Week 14.	Ayaan	8%	
5.0	**Documentation & Formatting Finalization**			**9%**
5.1	**5.0 References:** Compile References ensuring correct APA Style formatting.	Ayaan	5%	
5.2	**6.0 WBS & Final Formatting:** Finalize Work Breakdown Structure and ensure overall document formatting.	Zi Yee	4%	
Total			**100%**	